

THE PROBLEM

India does not have a national blood shortage; it has a coordination issue. Blood banks work alone, causing one hospital to face a serious shortage while another just 3 km away discards the same blood type because it has expired.

Case Study: The "Silent Wastage" Problem

While thousands of patients struggle to find donors, the absence of a real-time peer-to-peer network leads to significant waste. The following data shows how serious this disconnect is:

- **High Discard Rates:** Recent studies found that blood discard rates in Indian hospitals can reach up to 18%.
- **The Platelet Crisis:** Platelets, which have a very short shelf-life, account for 80-90% of all blood waste, mainly due to expiration rather than medical reasons.
- **Maternal Risk:** Research published in the Journal of South Asian Federation of Obstetrics and Gynaecology revealed that the lack of available blood is a key reason for surgical delays in maternal cases that are near-critical.
- **The "Golden Hour" Failure:** Even with storage facilities and databases, there is no last-minute way to transfer blood between nearby hospitals during emergencies, often resulting in preventable deaths.

Impacted Users: The Human Stakeholders

The status quo (a fragmented, manual system where hospitals operate as isolated silos) puts Medical Staff in the position of "Scarcity Triage," where surgery is driven by the contents of the fridge, not the needs of the patient. By creating a networked supply chain, we rebuild clinical confidence. For Patients and Families, fragmentation means a medical emergency becomes a logistical disaster, where they are forced to become "emergency couriers" while their loved ones are in surgery. Crimson makes logistics disappear, keeping families at the bedside.

Blood Bank Administrators are faced with a desperate gamble:

too much product means a tragedy of biological waste, while too little means a catastrophic loss of life. The need for predictive analysis is critical to ensuring product is used before it hits the bin. Lastly, the Donor Community experiences "Donor Fatigue" – a profound disillusionment when their "gift of life" is squandered because of a lack of communication between hospitals. Our aim is to honor that good will by ensuring every gift is meaningfully directed to the point of greatest impact.

THE SOLUTION - CRIMSON

The core of our app (**Crimson**) is to help hospitals never run out of blood by using a **peer to peer system of donation to ensure no one has excess or recess of blood - here the hospitals act as the donor and the receiver and hospital A can donate to hospital B**. Looking at the statistics of how many people who lost their lives due to the lack of a coordinated network of hospitals. Patients and hospitals can benefit from such a p2p network.

Key features :

Crimson's primary goal is to shrink the "Time-to-Transfuse" metric. By utilizing the **ARIMA** (AutoRegressive Integrated Moving Average) model, the app **analyzes a hospital's historical consumption patterns** like the Thursday night trauma surge or seasonal dengue spikes, to predict shortages before they happen.

Upon an unseen emergency, the app ensures you that you can communicate to every other hospital who is registered on this network within a 5 - 10 kilometer radius depending on the urgency of the request, thus **allowing a hospital with an emergency demand to get their supplies from a hospital who considers it as excess**, while at the same time not sending redundant requests to hospitals who can't help due to their presence in a farther locality

Beyond its primary purpose of cross hospital collaboration, Crimson also ensures hospitals and individual donors that their donation doesn't go in vain. If a person donates blood that already exists in excess – or if they belong to a blood group that doesn't face repeated shortages, they never go to waste because the app ensures it arrives in time

The Value-add:

It is to note that hospitals don't need to change their working procedure, we just need access to your data. The solution adds value to all the following functioning bodies.

Doctors: No more cancelled surgeries. No more telling a family, "We're waiting on the blood bank." Ensuring peace of mind that help is 10 kilometers away maximum.

Administrators: Automated audits and "Reciprocity Credits." When you share today, you earn "Priority Points" for when you're in a pinch tomorrow.

For the System: We turn a ₹500 Crore national loss into a Net-Zero Waste economy.

The Bottom Line:

Every expired unit of blood is a medical tragedy and a financial drain. Every cancelled surgery is a lost opportunity.

We are replacing the "Silo Crisis" with a Peer network - a living, breathing network that ensures no Indian life is lost simply because the blood was in the wrong fridge.

Existing solutions :

Government based E-Raktkosh is the nearest alternative to crimson, but E-Raktkosh is a static database of all inventories across India, managed under the hood of government. Hence again posing the same staticity threat where you cannot “instant order 5 units of blood”. As the network is too large and the app is mostly used for updating national database. It is redundant for actual emergencies. The reason why we are different is because we are closer to the hospital networks and it involves a network of hospitals that are close to the requester than accessing a database of national reserve of blood banks.

Predicting analysis further improves the situation by it single handedly dealing with any shortages unless an actual breakout happens.

What We offer is a 2 step protection

1. Predictive analysis that stops blood related shortages from happening by timely notification when reserve gets low.
2. In case of emergency beyond the analysis, the peer network ensures that no lives are lost because of an inventory shortage at a local level.

Trust Gap:

We know hospitals are often competitors. To make them collaborators, we've built three pillars of trust:

Reciprocity Credits: Think of this as a "Karma Ledger." When Hospital A shares surplus units, they earn Priority Points. This ensures that if they ever face a shortage, the network prioritizes them in return. It turns competition into a safety net.

Privacy First: Hospitals see available units within the network, but never the private patient data of their rivals. We share the resources, not the secrets.

Digital Handcuffs (Compliance): Every transfer generates an automated Digital Chain of Custody. It meets all NABH and drug controller guidelines, replacing hours of manual paperwork with a single, legal digital audit.

The True Cost of the "Silo Crisis" – True economic figures

The Hidden economic Drain: Right now, every expired unit of blood is more than a medical tragedy, it's a massive financial leak. Processing a single unit costs between ₹1,500 and ₹3,000. In a large hospital, a 5% expiry rate translates to a million rupees.

Demise of a savable patient : Every lost patient is a family of shattered dreams, if the reason for their demise is a preventable shortage, it turns from a medical inability to medical irresponsibility. We believe no one should sit in overabundance while a neighbour has a shortage. Hence the proposal of Crimson.

FEASIBILITY OF IMPLEMENTATION

1. Secure & Frictionless Onboarding

Hospitals can onboard in under 10 minutes using existing desktops or tablets. The platform is entirely cloud-based, requires no specialized hardware, and integrates seamlessly into current hospital workflows.

Security is enforced through a zero-trust credentialing model. Only government-authorized blood banks and NABH-accredited hospitals are permitted access. Registrations are verified through a combination of manual review and API-based validation against official state drug controller and NABH databases. Unverified entities remain fully locked out of the system.

2. Nationwide Scalability by Design

The system is engineered for India-scale deployment. It functions as a lightweight digital coordination layer rather than a physical supply chain. Whether deployed across five hospitals in a district or thousands nationwide, the operational requirements remain unchanged.

Scaling the platform requires only cloud resource expansion. There are no physical bottlenecks, proprietary devices, or regional dependencies, enabling immediate expansion from pilot deployments to a pan-India network.

3. Technology Backbone

The platform operates through a responsive mobile and web application ecosystem, ensuring universal accessibility. Data integrity and security are maintained through a robust PostgreSQL database architecture.

Predictive intelligence is driven by ARIMA (Auto-Regressive Integrated Moving Average) models, converting historical utilization data into forward-looking blood demand forecasts to minimize shortages and prevent wastage.